

Investigating the content-specificity and time course of attentional biases in mathematics anxiety: a dot-probe approach

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ABSTRACT

Mathematics anxiety (MA) is associated with intrusive thoughts and performance difficulties, yet its impact on attentional processes remains uncertain. Previous findings on attentional bias (AB) in MA have been inconsistent, and no studies have explicitly examined biases towards the negative internal dialogue and ruminations characteristic of MA. Across three experimental studies, we employed the dot-probe paradigm to investigate the content specificity and time course of AB in MA. Study 1 ($n=50$) examined attentional responses to math-related words. No AB was observed, although higher MA was linked to slower reaction times (RTs) on congruent trials. Study 2 ($n=83$) tested rumination-related words presented for 500ms and 1250ms. Again, no significant AB emerged, but high MA participants responded more slowly overall, independent of stimulus type or exposure duration. Study 3 ($n=64$) extended this by including math-unrelated negative words and neutral-neutral baseline trials to disentangle vigilance and disengagement components. No robust AB effects were found, although weak positive correlations appeared between MA and RTs in specific conditions. Overall, MA was consistently associated with slower RTs rather than selective attentional bias, suggesting a generalised reduction in processing efficiency. These findings challenge the assumption that MA is underpinned by a content-specific AB, instead indicating broader attentional inefficiency.

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Introduction

Math anxiety is defined as ‘a feeling of tension and anxiety that interferes with the manipulation of numbers and solving of mathematical problems in a wide range of ordinary life and academic situations’ (Richardson & Suinn, 1972). It affects students across all educational levels, from preschool (del-Río et al., 2023) to primary (Hill et al., 2016; Petronzi et al., 2019), secondary (Chinn, 2009; Hill et al., 2016), and even higher education (Baloglu & Kocak, 2006; Hunt et al., 2011). It also persists into adulthood (Hart & Ganley, 2019), often correlating with lower math performance (Namkung et al., 2019) and decreased interest in STEM careers (Ahmed, 2018). Identifying the factors contributing to the development and maintenance of math anxiety is therefore crucial, with one such factor being the role of attention.

Negative internal dialog and ruminations in math anxiety

Math anxiety is associated with intrusive, worrisome thoughts and ruminations. Hunsley (1987) used the Cognitive Intrusions Questionnaire (CIQ) developed by Sarason (1984) to investigate undergraduate students’ thoughts before and after statistics exams. They found that math anxiety accounted for 15% of

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the variance in negative internal dialogue, even after controlling for general test anxiety. Building on this, Hunt et al. (2014) utilised a modified version of the CIQ to retrospectively assess students' self-reported intrusive thoughts during math tasks. Their findings revealed that individuals with higher math anxiety reported more frequent task-irrelevant intrusive thoughts. These thoughts included concerns about time pressure, potential judgement from others, panic, physical symptoms (such as sweating and rapid heart rate), past negative experiences with math, and fear of making mistakes. Notably, thoughts related to time pressure occurred even in the absence of actual time constraints. Moreover, Hunt *et al.* discovered that the extent to which these intrusive thoughts were perceived to hinder performance mediated the relationship between math anxiety and actual performance. This mediation was particularly evident in tasks that demanded greater working memory resources. These findings underscore the critical role of intrusive thoughts and ruminations in math anxiety and in explaining the impact of math anxiety on performance.

Attentional bias in math anxiety

According to Attentional Control Theory (Eysenck et al., 2007), anxiety affects an individual's attentional control, causing an imbalance between the stimulus-driven (bottom-up) and the goal-directed (top-down) attentional systems, which would make anxious individuals more vulnerable to bottom-up attentional intrusions, including the negative internal dialog and ruminations described above. The stronger influence of the stimulus-driven attentional system in individuals with high anxiety is believed to underlie their tendency to focus more on threat-related stimuli than on neutral ones, leading to a bias in attention toward threatening information (Eysenck et al., 2007; Eysenck & Byrne, 1992), that is a characteristic of several anxiety disorders (Bar-Haim et al., 2007). Research has shown that high math-anxious individuals display an AB toward math-related information (Rubinsten et al., 2015; Suárez-Pellicioni et al., 2015). AB is characterised by three components: vigilance or engagement (i.e. the initial focus on a stimulus), disengagement (i.e. the ability to shift focus away from a stimulus), and avoidance (i.e. actively steering attention away from a stimulus). Research indicates that anxiety increases vigilance towards threatening information (e.g. engagement) and contributes to difficulties disengaging from it (Buckner et al., 2010; Leleu et al., 2014). Avoidant behaviours have also been described, where individuals attempt to evade the anxiety-inducing stimulus altogether (Koster et al., 2006).

While significant research has explored attentional processes in various forms of psychopathology, findings on AB related to math anxiety have been inconsistent. Hopko et al. (2002) used a computer-based version of an Emotional Stroop task in which participants are presented with threatening (i.e. math-related) and neutral words displayed in different colours and the participants' task is to indicate the colour of the ink (Williams et al., 1996). Slower responses are expected to threatening words as compared to neutral ones, which is considered to indicate the allocation of attention to the processing of the threatening word content instead of solving the main task of reporting the words' ink color. They found no significant performance differences based on math anxiety level (high vs. low) or word type (mathematical vs. neutral), which they attributed to having used math-related words that were too abstract (e.g. polynomial, theorem), so high math anxious individuals would be less familiar with them. Using the same approach, Suárez-Pellicioni et al. (2015) observed that individuals with high math anxiety exhibited delayed responses to reporting the ink colour of math-related words, exhibiting a greater AB compared to those with lower anxiety. The emotional Stroop task, however, has been criticised for measuring a general threat-related slowdown rather than selective attention mechanisms (Algom et al., 2004). Conversely, the dot-probe detection task (also referred to as the visual probe task) is a more precise method for identifying the subcomponents of AB (MacLeod et al., 1986). The dot-probe task involves presenting two competing stimuli (e.g. math-related and neutral word pairs) simultaneously on a screen. After a set period, these stimuli are replaced by a visual probe (e.g. '*'), and participants must quickly indicate the probe's location (i.e. left or right). Using a computerized numerical version of the dot probe task, Rubinsten et al. (2015) found that high math anxious individuals reacted faster when the probe was presented at the location of the numerical related stimuli (congruent trials), suggesting vigilance to threat and supporting the existence of AB in math anxiety. Further, Pizzie and Kraemer (2017) utilised fMRI and a dot-probe task to investigate high math anxious individuals' response to briefly presented

negative images (e.g. a bleeding hand) and mathematical expressions (e.g. $y=2x\sqrt{1-y^2}$). They found that high math anxious individuals showed greater amygdala activation to math stimuli, indicating a threat response. Behaviourally, however, they found avoidance that was specific of math stimuli. This study therefore showed evidence supporting an initial vigilance for math-related stimuli followed by their avoidance. While not being part of AB towards threat, shifting efficiency, referring to the ease with which attention is redirected, has also been shown to be impacted by math anxiety (González-Gómez et al., 2023). Together, these mixed findings suggest that while attentional bias may play a role in math anxiety, its manifestation appears to depend on the specific cognitive task used. Further research using methodologically diverse approaches are needed to clarify the components of AB affected by math anxiety.

The importance of stimulus timing in understanding the components of AB

Studies on attentional processes in math anxiety often rely on Attentional Control Theory (ACT, Eysenck et al., 2007) which builds on the earlier Processing Efficiency Theory (PET, Eysenck & Calvo, 1992). As mentioned earlier, ACT claims that anxiety creates an imbalance between the stimulus-driven (bottom-up) and the goal-directed (top-down) attentional systems, causing individuals to pay attention to irrelevant internal or external stimuli rather than focusing on the task at hand. The Vigilance-Avoidance Hypothesis (Derakshan et al., 2007), on the other hand, suggests that the attentional bias is dynamic and manifests as initial heightened vigilance towards threatening cues (e.g. math-related words), followed by avoidance of these cues. Previous studies have struggled to disentangle the different components of attentional bias due to their reliance on a single, relatively long stimulus presentation time (e.g. 1000 ms, Rubinsten et al., 2015), which allows multiple shifts of attention and obscures its temporal dynamics. To fully understand the dynamics proposed by the Vigilance-Avoidance Hypothesis, it is essential to manipulate stimulus presentation time, presenting stimuli for a short duration (e.g. 100–200 ms) to capture initial vigilance and for longer durations (e.g. 500–1500 ms) to detect later avoidance tendencies.

The importance of the type of content in determining AB

Most research in this area has focused on ABs toward numerical or math-related stimuli, such as math equations (e.g. $26+65$, Rubinsten et al., 2015) or math-related words (e.g. 'quantity' or 'equation', Rubinsten et al., 2015; Suárez-Pellicioni et al., 2015). However, to date, it remains unclear whether individuals with high math anxiety exhibit AB toward stimuli that represent the negative internal dialogue and ruminations characteristics of math anxiety (Colomé et al., 2023; Rubinsten et al., 2012), such as worry, self-doubt, and catastrophic thinking. Rubinsten et al. (2012) used an affective priming task to explore implicit emotional associations with math. Participants were asked to verify arithmetic equations as correct or incorrect, with each equation being preceded by a prime word - positive, neutral, negative, or math-related. Results showed that males responded faster when equations were preceded by positive primes, while females responded faster following negative primes. Notably, females also responded to math-related words in a manner similar to negative primes, suggesting a negative emotional association with math that was not observed in males. Although the study did not directly measure attentional bias, the priming effects suggest that some women may be more attuned to threat-related cues in math contexts. This aligns with work by Shi and Liu (2016), who found that math-related sentences evoking dysfunctional thoughts, such as '*I feel nervous working on a math problem*', impaired working memory in high math-anxious individuals. This indicates that external cues linked to negative internal thoughts can significantly impact cognitive and attentional processing, reinforcing the need to examine attentional biases not only toward math-related content but also toward rumination-related content.

Taken together, to the best of our knowledge, no study has examined whether the intrusive and ruminative cognitions characteristic of math anxiety function as attentional bias stimuli. This represents an important gap in the literature, as Attentional Control Theory (Eysenck et al., 2007) posits that anxiety reduces goal-directed attentional control, leaving individuals vulnerable not only to external threats but also to internally generated intrusive thoughts. Rumination may therefore act as an internal threat cue, competing for limited cognitive resources and amplifying the negative effects of math anxiety on performance. By including rumination-related stimuli alongside math-related and general negative stimuli,

the present research seeks to provide a more comprehensive theoretical account of attentional processes in math anxiety, and to test whether intrusive cognitions elicit vigilance, disengagement difficulties, or avoidance in a manner comparable to external math-related threats.

Dot probe task

This paper presents three studies investigating attentional processes in math anxiety using a dot-probe paradigm. This task comprises two conditions. In the congruent condition, the visual probe appears in the same spatial location as the math-related (threatening) stimulus. In contrast, in the incongruent condition, the probe appears in the location previously occupied by the neutral stimulus. Faster response times on congruent trials are interpreted as evidence of attentional vigilance toward threat-relevant stimuli, suggesting that participants' attention was already oriented toward the location of the threatening word, thereby facilitating quicker detection of the probe. Conversely, slower responses on incongruent trials may reflect difficulty disengaging from threat, indicating that attention remained fixated on the threatening stimulus, making slower the detection of the probe on the opposite side of the screen. Finally, faster responses on incongruent trials may be indicative of attentional avoidance, whereby participants actively divert attention away from the threatening stimulus, resulting in enhanced detection of probes appearing in the neutral location.

Study 1: Is math anxiety associated with AB towards math-related words? (replication of previous studies)

Study 1 investigated whether individuals with high or low math anxiety exhibit an AB towards math-related words. We hypothesised that individuals with high math anxiety would show an AB towards math-related words, replicating previous research (Rubinsten et al., 2015; Suárez-Pellicioni et al., 2015). This would be indicated by vigilance towards math stimuli (faster response times when identifying probes that replaced math-related words), difficulty disengaging from math stimuli (slower response times when identifying probes that replaced non-math-related words), or avoidance of math stimuli (faster responses to probes replacing non-math related words). Additionally, we predicted a correlation between math anxiety and response times on incongruent trials: either a negative correlation, where higher math anxiety leads to slower response times due to difficulty disengaging from math-related stimuli, or a positive correlation, where higher math anxiety leads to faster response times due to avoidance of the math-related stimuli (c.f. Pizzie & Kraemer, 2017; Colomé et al., 2023; Van den Bussche et al., 2020).

Study 2: Is math anxiety associated with AB towards rumination-related words?

The aim of Study 2 was to investigate AB towards rumination-related words (e.g. 'Panic', 'Worry', etc) associated with MA. The inclusion of these stimuli represents a novel contribution, as prior research on MA has primarily examined AB towards external math-related stimuli (e.g. math-related words), with no attention to intrusive, ruminative cognitions that characterise this anxiety. ACT (Eysenck et al., 2007) proposes that anxiety reduces goal-directed attentional control, making individuals susceptible to stimulus-driven intrusions. Rumination can therefore be conceptualised as an internally generated analogue of threat-related cues, competing for limited attentional resources and impairing performance. Thus, by testing AB towards rumination-related words, Study 2 extends prior work (Rubinsten et al., 2015) to examine whether these intrusive cognitions elicit vigilance, difficulty disengaging or avoidance, thereby offering a more comprehensive account of the cognitive mechanisms underlying MA.

To better understand the different AB components, this study employed two exposure durations to measuring early attentional processes such as vigilance (i.e. 500ms) and later attentional processes such as avoidance or difficulty in disengagement (i.e. 1250ms) (Gaffiero et al., 2019; Mogg et al., 1997). We hypothesised that higher math anxiety would be linked to increased vigilance towards math-rumination words during the 500ms exposure period, reflected by faster response times to probes in congruent trials. This prediction aligns with previous research suggesting that anxiety heightens vigilance within brief exposure times (Klumpp & Amir, 2009; Mogg et al., 2004). For the 1250ms exposure duration, we

did not have strong predictions, as previous studies have shown evidence compatible with HMA individual's difficulty disengaging from math (Rubinsten et al., 2015) and evidence of avoidance (Pizzie & Kraemer, 2017). Slower response times to incongruent stimuli would be interpreted as evidence supporting difficulty disengaging from rumination-related words whereas faster response times to incongruent stimuli would be interpreted as evidence supporting avoidance of these stimuli.

Study 3: Is math anxiety associated with AB towards math-unrelated negative words?

Study 3 included a set of negative words unrelated to math (e.g. 'Fight', 'Evil', etc), aiming to determine whether individuals with high math anxiety exhibit an AB that goes beyond math-related stimuli and generalises to math-unrelated negative stimuli. This study employed the stimuli duration in Study 2 to explore early and later components of AB toward math-unrelated negative stimuli. We did not expect high math anxious individuals to show AB toward these math-unrelated negative words, regardless of stimulus duration. To our knowledge, this is the first study to examine this question using a dot-probe task (see MacLeod et al., 1986) for a similar approach with images). While Rubinsten et al. (2015) included negative words in a priming paradigm, and most recently Cuder et al. (2025) used a dot-probe task with math and neutral words, no previous research has tested whether math anxiety is associated with AB toward general negative stimuli using a methodological approach that distinguishes between vigilance, difficulty disengaging and avoidance.

Study 1

Materials and methods

Design

Study 1 employed a mixed-measures experimental design. The first independent variable, math anxiety, was a between-subjects factor with two levels: high and low, determined by a median split of participants' scores in the Mathematics Anxiety Scale UK (Hunt et al., 2011). The second independent variable, probe congruency, was a within-subjects factor also with two levels: congruent and incongruent. The dependent variable was response time to correctly solved trials, recorded in milliseconds (ms).

Participants

The sample comprised 50 participants, including 8 males (16%, $M \pm SD$ years, 23.63 ± 8.28) and 42 females (84%, $M \pm SD$ years, 25.6 ± 9.31). All participants were undergraduate Psychology students at the University of Derby, located in the East Midlands of the U.K. The study was conducted as part of an undergraduate dissertation project at the University of Derby. In line with institutional policy at the time, undergraduate projects were reviewed through the School of Psychology ethics review process rather than being assigned individual ethics reference numbers. Ethical approval for this study was therefore granted via the School of Psychology 'Independent Study in Psychology' module review procedure, which complies with institutional and British Psychological Society ethical guidelines.

To be eligible, participants had to be: i) at least 18 years old; ii) have corrected or corrected-to-normal vision; and iii) be in good mental health, with no current diagnosis of a mental health disorder or learning disorders. Recruitment was conducted through opportunity sampling, using advertisements and leaflets to attract participants from the general university population. In exchange for their participation, individuals were either awarded research participation scheme points or entered a prize draw for a selection of books with an approximate cost of £100.

Materials

Demographic questionnaire

A demographic questionnaire was used to record participants' age and sex.

Mathematics anxiety scale UK (MAS-UK)

The MAS-UK (Hunt et al., 2011) was used to assess how anxious participants would feel in a variety of situations concerning math, such as taking a math exam. The MAS-UK is a 23-item self-report instrument that uses a five-point Likert scale ranging from 1 ('Not at all') to 5 ('Very much so'). Total scores range from 23 to 115, with higher scores indicating a higher level of math anxiety. The scale possesses excellent internal consistency ($\alpha = 0.96$) and excellent test-retest reliability ($\alpha = 0.89$) (Firouzian et al., 2015; Hunt et al., 2014). Math anxiety scores across the entire sample had a mean of 53.26 ± 18.93 , with an interquartile range (IQR) of 23.25. The scores ranged from 26 to 109, reflecting a broad spectrum of math anxiety levels among participants.

State trait anxiety Inventory (STAI)

The STAI, developed by Spielberger and Gorsuch (Spielberger et al., 1983), was employed to measure state anxiety and trait anxiety. The STAI comprises 40 items, with 20 items assessing state anxiety (how individuals feel at a specific moment e.g. calm) and 20 items evaluating trait anxiety (how individuals generally feel e.g. confident). This scale uses a four-point Likert scale ranging from 1 ('Not at all') to 4 ('Very much so'), while the trait anxiety items use a scale from 1 ('Almost never') to 4 ('Almost always'). Reliability analyses show high internal consistency for the state anxiety component ($\alpha = 0.92$) and for the trait anxiety component ($\alpha = 0.90$). Additionally, the test-retest reliability has been reported as $\alpha = 0.86$ (Barnes et al., 2002; Kabacoff et al., 1997).

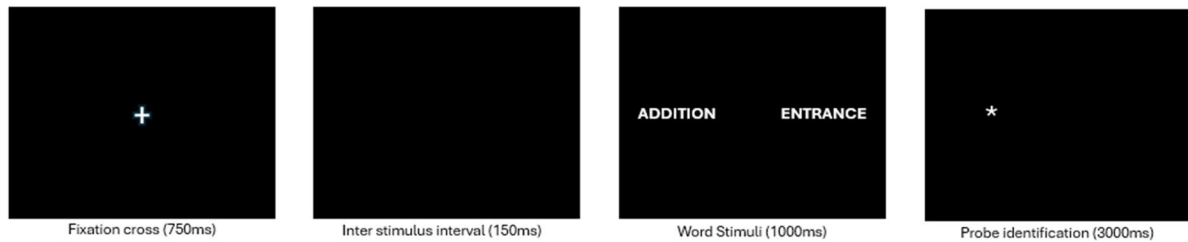
Experimental task and stimuli

AB was assessed using a dot-probe task designed in Millisecond Inquisit (software version 4.05.0). Participants were seated approximately 60cm from a Samsung R780 17.3-inch laptop screen. In each trial, two prime stimuli - a math word (e.g. addition, counting, division) and a neutral word (e.g. entrance, keyboard, notebook) - were presented horizontally on the screen, 13.5cm apart, for a fixed duration (1000ms). Following the presentation of the word pair, a visual probe (an asterisk) appeared at the location of one of the former stimuli: either the math word (congruent) or the neutral word (incongruent). Participants were required to indicate the location of the probe (left or right) by pressing one of two buttons on a Cedrus RB-834 response pad (red=left, blue=right).

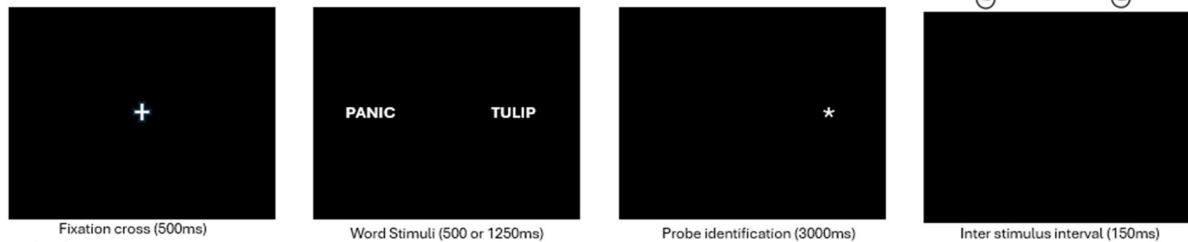
The dot-probe task measured participants' accuracy and response times (RT) for identifying the visual probe when it replaced a math-related word and when it replaced a neutral word. Each trial began with a fixation point ('+') displayed at the centre of the screen for 750 milliseconds, followed by an inter-stimulus interval of 150 milliseconds. Then, the pair of words was shown for 1000 milliseconds. After the words disappeared, a visual probe ('*') appeared on either the left or right side of the screen, in the location of one of the previously displayed words. The probe remained on screen for up to 3000 milliseconds, during which participants were instructed to indicate the location of the probe by pressing the corresponding red (left) or blue (right) button on the response pad. If no response was made within 3000 milliseconds, the trial ended automatically and the next trial began. Please see Figure 1 showing stimuli presentation and their timing for the three studies. Note: In Studies 2 and 3, the early ISI following fixation (prior to word presentation) was removed and reallocated to occur after the participant's response. Although the total ISI duration remained constant across studies 1–3, this adjustment served to better support sustained attention and response quality over a longer task. Specifically, placing the ISI post-response allowed participants a brief cognitive pause between trials, which helped mitigate fatigue and reduce speed-accuracy trade-offs in the single-block design with a higher trial count. This shift maintained temporal consistency while optimising the cognitive pacing of the task.

The stimuli included 16 math words (e.g. Addition, Division, Multiply) and 16 neutral words (Entrance, Table, Paper). To control for potential confounding variables, all words were of equal length (8 letters). Most of the words were sourced from previous research that selected math and neutral words based on their frequency and emotional load (Rubinsten et al., 2015). Additional words were selected based on their length and familiarity. Each word pair was displayed against a black background, in capital letters, using a white Arial 24-point font. To ensure variability, each math word was paired with a different

Study 1



Study 2



Study 3

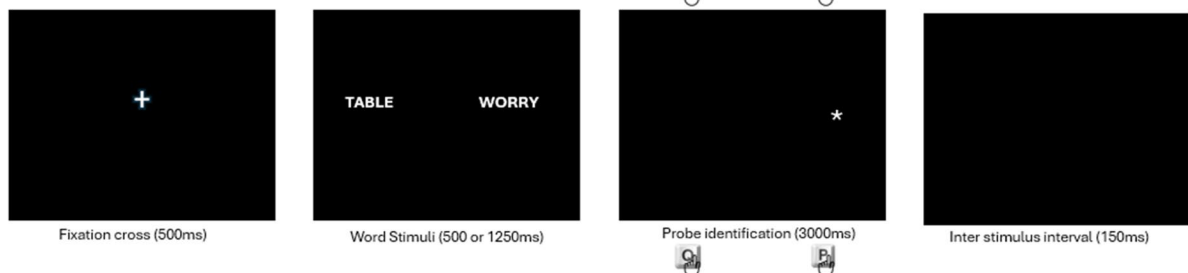


Figure 1. Pictorial representation of the dot-probe task trial events for studies 1, 2 and 3.

neutral word, and no word appeared more than twice on one side of the screen. The location of the probe was randomised between the left and right sides of the screen for each trial. However, the programme was designed to ensure that, by the end of the experiment, the probe appeared on each side an equal number of times. This randomisation was implemented to prevent participants from developing a bias towards one side of the screen. The task began with 10 practice trials to familiarise participants with the procedure (using different stimuli). The main task involved presenting the 16-word pairs four times each, for each of the four possible word-probe combinations (i.e. left and right word location and left and right target probe location), resulting in a total of 74 trials.

Procedure

Participants were invited to participate individually, in a single laboratory session, located at the University of Derby. Upon procurement of informed consent (written), participants completed the demographic questionnaire, STAI and MAS-UK scales. Next, participants completed the dot-probe task. Upon completion, participants were debriefed and thanked for their time. Participation lasted approximately 20 minutes.

Results

All participants who attended the laboratory session completed the measures in full; therefore, there were no missing data. Data screening indicated that parametric assumptions were broadly met. All variables met checks for ZSkew/ZKurtosis, the Shapiro–Wilk test of normality, and Z-scores, except for the STAI. The STAI exceeded the ± 1.96 threshold for ZSkew, was significant in the Shapiro–Wilk test ($p < 0.001$) and had one outlier. Inspection revealed a score of 76, which is within the possible range of 20–80. Although extreme, this value is legitimate and reflects the upper end of the anxiety continuum. Given that parametric analyses are robust to moderate violations of normality and the outlier was valid, the STAI was retained in the analyses.

Table 1. Mean (M) response times (in milliseconds) and standard deviations (SD) for low and high math anxiety groups across conditions.

Condition	Low math anxiety group (M±SD)	High math anxiety group (M±SD)
Congruent	311.54±43.70	325.59±39.87
Incongruent	311.26±48.96	327.75±46.51

Accuracy in the dot-probe task was very high for congruent ($M \pm SD$; 99.06%±0.019%) and incongruent trials ($M \pm SD$; 99.06%±0.018%). This is expected based on the simplicity of the task. An independent samples t-test was used to compare math anxiety between the high and low math anxiety groups, $t(48) = -8.015$, $p < 0.001$ (95%, CI, LCI: -35.47, UCL: -21.24). Math anxiety was significantly higher in the high (67.44 ± 16.60) compared to the low (39.08 ± 6.13) math anxiety group. Independent samples t-tests were also performed to compare state and trait anxiety between the groups, revealing no significant differences in state, $t(48) = -1.478$, $p = 0.073$ (95%, CI, LCI: -12.37, UCL: -1.90), or trait anxiety, $t(48) = -0.922$, $p = 0.361$ (95%, CI, LCI: -9.80, UCL: -3.64).

A mixed measures ANOVA was then used to compare RT between congruent and incongruent trials by math anxiety group. No significant main effect of probe congruency, $F(1, 48) = 0.128$, $p = 0.722$, math anxiety, $F(1, 48) = 1.510$, $p = 0.225$ or interaction, $F(1, 48) = 0.218$, $p = 0.643$ was observed. For descriptive statistics please see Table 1 below.

To further investigate potential differences between math anxiety groups, a quartile split approach was employed, with analyses focusing on the upper and lower quartiles - that is, the 13 participants with the highest and the 13 with the lowest math anxiety scores. This method was chosen to maximise the contrast between individuals with markedly different levels of math anxiety, thereby increasing the sensitivity to detect group differences that may be obscured by variability in the full sample. Independent-samples t-tests were conducted to compare performance between the high and low math anxiety groups across congruent and incongruent conditions. However, no statistically significant differences in RT were observed between groups for either the congruent ($p = 0.07$) or incongruent ($p = 0.162$) trials.

Bivariate correlations were then used to examine the relationship between math anxiety (using the raw MAS-UK score) and RT for both congruent and incongruent trials. A significant positive correlation was found between math anxiety and mean RT for congruent trials, $r(48) = 0.29$, $p = 0.02$, please see Figure 2 below. This indicated that, as math anxiety increased, participants took longer to respond on trials where the probe replaced math-related words. However, no significant relationship was observed between math anxiety and mean RT for incongruent trials, $r(48) = 0.22$, $p = 0.06$.

An AB index was then created by subtracting the mean RT for congruent trials from the mean RT for incongruent trials. In this index, positive scores indicate an AB towards math-related stimuli, neutral scores suggest no AB, and negative scores reflect an AB towards neutral stimuli. There was no significant bivariate correlation between math anxiety and AB index, $r(48) = 0.08$, $p = 0.28$.

Study 2

Materials and methods

Design

A mixed-measures experimental design was used with three independent variables. Math anxiety (high vs. low) was the between-subjects factor, probe congruency (congruent vs. incongruent) and presentation duration (short: 500ms, long: 1250ms) were the within-subjects factors. The dependent variable was RT for correctly solved trials, measured in milliseconds (ms).

Participants

Participants ($N = 85$) were recruited via the same method as Study 1. The sample comprised 18 males (24.50 ± 6.39) and 67 females (28.46 ± 10.04). No significant differences were observed in age ($p = 0.124$) or sex ($p = 0.117$) between the two groups. Eligibility criteria stated: i) at least 18 years old; ii) corrected

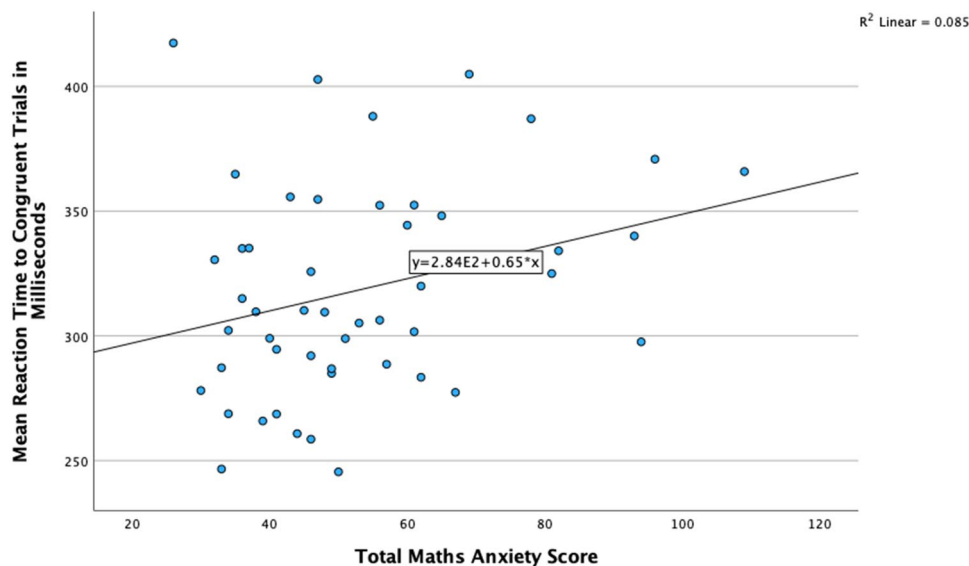


Figure 2. Positive linear relationship between total maths anxiety scores and mean response time to congruent trials on the dot-probe task.

or corrected-to-normal vision, and iii) no current diagnosis of a mental health disorder (e.g. anxiety) or learning disorders. Participants were recruited through various social media platforms including X (formerly Twitter) and Facebook. No incentives were offered to participants, except for research participation points, if the participant was a university student. Ethics approval number ETH2021-2024.

Materials

Consistent with Study 1, participants were provided with a demographic questionnaire (age, sex) and the MAS-UK scale (Hunt et al., 2011) to measure participants' level of math anxiety. The math anxiety scores for the entire sample in Study 2 had a mean of 53.14 ± 18.38 , with a median of 50. Scores ranged from 24 to 105, with an interquartile range (IQR) of 30, indicating a moderate spread of scores. Study 2 also measured test anxiety using the Westside Test Anxiety Scale (WTAS) and general anxiety using the Beck Anxiety Inventory (BAI).

Westside test anxiety scale (WTAS)

The WTAS is a widely used instrument designed to assess test anxiety in individuals. Developed by the Westside Consortium (1997), this scale consists of 10 items that measure the intensity of test-related anxiety experienced by individuals. The WTAS is specifically tailored to capture anxiety related to academic testing and performance situations. Each item on the WTAS is rated on a 5-point Likert scale ranging from 1 ('Not at all') to 5 ('Very much'), reflecting the degree to which the respondent experiences anxiety in testing situations. The scale's items address various aspects of test anxiety, including worry, fear of failure, and physiological responses. Scores range from 10 to 50, with higher scores indicating higher levels of test anxiety. The scale possesses good internal consistency, with a composite reliability value of 0.88 (Talwar, 2019).

Beck anxiety inventory (BAI)

The BAI is a well-established self-report tool designed to measure the severity of anxiety symptoms. Developed by Beck et al. (1988), the BAI comprises 21 items that evaluate the intensity of various symptoms associated with anxiety, including physiological and cognitive aspects. Respondents rate each item on a 4-point scale from 0 ('Not at all') to 3 ('Severely – it bothered me a lot'), indicating how much they

have been bothered by each symptom over the past week. The BAI covers a broad range of anxiety symptoms such as nervousness, dizziness, and numbness, providing a comprehensive assessment of anxiety levels. The total score is obtained by summing the ratings for all 21 items, with possible scores ranging from 0 to 63. Higher scores indicate higher levels of anxiety. The English version of the BAI possesses excellent ($\alpha=0.91$) internal consistency and moderate (0.65) test-retest reliability (Bardhoshi et al., 2016).

Experimental task and stimuli

The dot-probe task was programmed using Millisecond Inquisit (software version 4.05.0). Participants were instructed to place their left and right index fingers on the 'Q' and 'P' keys of a standard computer keyboard, and to respond by pressing the corresponding key (Q=left, P=right) as soon as the visual probe (*) appeared. Each trial began with a fixation cross ('+') displayed at the centre of the computer screen for 500ms, followed by a word pair presented simultaneously for either 500ms or 1250ms. Immediately after the word pair disappeared, a visual probe ('*') appeared in the location of one former stimulus, remaining on screen for a maximum of 2000ms. If no response was made within this time, the probe was removed, and after a brief inter-stimulus interval of 150ms, the next trial began. Response time was only measured for trials in which visual probes were correctly identified. The order of trials was randomised for each participant. Prior to the main task, participants completed eight practice trials featuring neutral stimuli to familiarise themselves with the task requirements.

The task included 16-word pairs, each consisting of one rumination-related word and one neutral word. In total, there were 128 experimental trials, balanced across conditions of probe congruency (congruent vs. incongruent) and presentation duration (500ms vs. 1250ms, mixed in the same block), ensuring each word pair appeared under each condition. The rumination-related words were chosen to reflect common intrusive thoughts associated with math anxiety, as identified by previous research (Hunsley, 1987; Hunt et al., 2014), and included words like 'Panic', 'Test', and 'Worry'. Neutral words, such as 'Tulip', 'Shoe', and 'Table', were matched with rumination-related words for letter count (6.44 ± 2.22), ensuring consistency in visual presentation. Rumination-related and neutral words were matched on emotional properties, including valence and arousal, using data from the Affective Norms for English Words database (Bradley & Lang, 1999).

Procedure

Participants were invited to participate in an online computerised task using Qualtrics (Provo, UT). Following informed consent (written), and to minimise demand characteristics, participants completed the dot-probe task followed by the demographic questionnaire, MAS-UK scale, WTAS, and BAI. Upon completion, participants were debriefed and thanked for their time. Participation lasted approximately 30 minutes.

Results

Response time data were screened for implausible responses (< 200 ms and > 2000 ms). Two participants were excluded on this basis, leaving a final sample of 83. Further checks revealed no significant outliers. While some variables showed minor deviations from normality (e.g. skewness and Shapiro-Wilk significance), these were not severe, and parametric analyses were deemed appropriate given their robustness to moderate non-normality.

To examine the effects of math anxiety, probe congruency and presentation length on RT, a 2 (math anxiety: high vs. low) $\times$ 2 (probe congruency: congruent vs. incongruent) $\times$ 2 (presentation length: short [500ms] vs. long [1250ms]) mixed ANOVA was performed.

The analysis revealed a significant main effect of math anxiety, $F(1, 81) = 7.371$, $p=0.008$, $\eta^2 = 0.083$, indicating that participants with high math anxiety responded more slowly overall compared to those with

Table 2. Mean (M) response times (in milliseconds) and standard deviations (SD) for low and high math anxiety groups across conditions.

Condition	Low math anxiety group (M ± SD)	High math anxiety group (M ± SD)
Congruent short	364.48 ± 58.67	402.17 ± 89.11
Incongruent short	361.11 ± 57.26	400.82 ± 85.54
Congruent long	337.24 ± 49.83	378.38 ± 88.39
Incongruent long	343.74 ± 51.35	386.52 ± 87.07

Participants with high math anxiety demonstrated significantly slower response times compared to those with low math anxiety.

low math anxiety, regardless of stimulus type (congruency) and stimulus duration (please see Table 2 below). This pattern indicates that high math anxiety is associated with generally slower processing of certain stimuli.

Additionally, there was a significant main effect of probe congruency, $F(1, 81) = 27.569$, $p < 0.001$, $\eta^2 = 0.254$, suggesting that participants, regardless of anxiety level, responded faster on congruent trials than on incongruent trials. No significant differences were observed in presentation duration, $F(1, 81) = 0.519$, $p = 0.474$, $\eta^2 = .006$. Moreover, no significant interactions were observed between math anxiety, probe congruency and presentation length (all $p > 0.05$).

Further analysis using bivariate correlations revealed significant, weak, positive correlations between math anxiety and mean RT for all trial types: congruent short trials, $r(83) = 0.29$, $p = 0.005$, incongruent short trials, $r(83) = 0.273$, $p = 0.006$, congruent long trials, $r(83) = 0.327$, $p = 0.001$ and incongruent long trials, $r(83) = 0.382$, $p < .001$. This indicates that higher levels of math anxiety were associated with longer RT in all conditions.

Lastly, attentional bias indices (incongruent vs. congruent) were calculated separately for short and long presentation trials, and their correlations with math anxiety scores were examined. The results revealed no significant relationship between math anxiety and attentional bias (AB) index scores in either the short trials, $r(83) = 0.084$, $p = 0.226$, or the long trials, $r(83) = -0.047$, $p = .337$. These findings suggest that math anxiety was not associated with the degree of attentional bias toward math-related stimuli under either temporal condition.

Study 3

Materials and methods

Design

A 4-way mixed-measures experimental design was employed, featuring math anxiety group (high vs. low) as the between-subjects factor and three within-subjects factors: word content (rumination-related, math-unrelated-negative, and neutral), probe congruency (congruent vs. incongruent), and presentation duration (500 ms vs. 1250 ms).

Participants

The sample comprised 111 participants, recruited via opportunity sampling via the distribution of study advertisements on X (formerly Twitter), Facebook and Reddit. Eligibility criteria stated: i) aged 18+, ii) normal or corrected-to-normal vision, and iii) no history of anxiety and/or depression or learning disorders. Forty-seven participants were removed due to incomplete responses; thus, the final sample included 64 participants. There were 13 males (30.85 ± 13.36) and 50 females (30.28 ± 10.83), and 1 participant who preferred to not declare their sex. 49.2% of participants were white, 66.2% British, 83.1% cited 'English' as their first language and 60% were pursuing a higher education level qualification. No significant differences were observed in age ($p = 0.295$) or sex ($p = 0.164$) between the two math anxiety groups. Ethics approval number = ETH2122-3192.

Materials

Consistent with Study 2, participants completed the MAS-UK scale, WTAS and BAI. The demographic questionnaire was expanded to include questions pertaining to age, sex, ethnicity and education level. The distribution of math anxiety scores across the entire sample revealed a broad range of anxiety levels, with a mean score of 55.31 ± 19.90 and a median score of 52.5. The scores ranged from 25 to 99, with an interquartile range (IQR) of 33, indicating a moderate spread around the central values.

Experimental task and stimuli

The experimental task was largely consistent with Study 2 and conducted online using Qualtrics (Provo, UT) software due to the global COVID-19 pandemic resulting in the suspension of face-to-face participant recruitment. A set of math-unrelated more general negative words, including words such as 'Fight', 'Evil', and 'Angry', were presented to participants to determine whether those with high math anxiety would display AB toward rumination-related words and math-unrelated negative stimuli. Each rumination-related and math-unrelated negative word was paired with a neutral word. All word pairs were matched for the number of letters. Additionally, a set of neutral-neutral word pairs were included to provide a measure of baseline attention, and a reference point for how quickly participants respond when no threat is present. The absence of this baseline condition has hindered previous studies from disentangling the effects of vigilance from those of difficulty disengaging from threat (Rubinsten et al., 2015). This is crucial for the correct interpretation of the AB index described above (incongruent vs. congruent), as a positive AB index could mean faster responses to congruent trials (vigilance), slower responses to incongruent trials (difficulty disengaging) or both. By including a baseline condition, we can understand which of the components is driving the effects, as vigilance is inferred when RTs are faster for congruent trials compared to baseline trials and difficulty disengaging is inferred when RTs are slower for incongruent trials compared to baseline trials (Koster et al., 2004). Consistent with Study 2, the rumination-related and neutral words were also matched on emotional properties, including valence and arousal.

A total of 48-word pairs were used in the task, comprising 16 rumination-related/neutral pairs, 16 math-unrelated-negative/neutral pairs, and 16 neutral-neutral baseline pairs. Before starting the main task, participants completed 8 practice trials using neutral stimuli not included in the main task to familiarise themselves with the task requirements. The main task consisted of 384 trials, with each word pair (rumination-related/neutral, math-unrelated-negative/neutral, and neutral-neutral baseline) presented four times. This repetition accounted for the two levels of probe congruency (congruent and incongruent) and two presentation lengths (500ms and 1250ms). The trials were structured to counterbalance the location of the word pairs (left or right) and the target probe location (left or right) across these conditions, ensuring an even distribution and preventing bias.

Procedure

Following informed consent (written), participants completed the dot-probe task, then they completed the demographic questionnaire, MAS-UK scale, WTAS, and BAI before being debriefed and thanked for their time. The full study lasted approximately 50 minutes.

Results

Data were screened for normality and implausible RTs. One participant was excluded as a significant outlier. The remaining RTs ranged from 229–942ms, which is within the acceptable response window. Therefore, no further trimming was required. Given that parametric tests are robust to moderate departures from normality with samples of this size ($n=63$), parametric analyses were deemed appropriate. Accuracy in the dot-probe task was high ($M \pm SD$; $96\% \pm 0.004\%$).

A 2 (math anxiety group; high vs. low) $\times$ 2 (probe congruency; congruent vs. incongruent) $\times$ 2 (presentation length; 500ms vs. 1250ms) $\times$ 3 (stimulus type: rumination-neutral, math-unrelated negative-neutral, and neutral-neutral) mixed ANOVA was conducted. No significant main effects of math

Table 3. Mean (M) response times (in milliseconds) and standard deviations (SD) for low and high math anxiety groups across conditions.

Condition	Low math anxiety group (M ± SD)	High math anxiety group (M ± SD)
Congruent short maths	384.61 ± 68.05	414.72 ± 139.19
Incongruent short maths	373.08 ± 64.90	390.47 ± 96.46
Congruent long maths	364.63 ± 72.89	385.34 ± 131.26
Incongruent long maths	361.17 ± 72.68	382.01 ± 91.80
Congruent short negative	372.55 ± 73.27	407.08 ± 116.53
Incongruent short negative	373.01 ± 65.43	393.23 ± 105.59
Congruent long negative	365.09 ± 77.81	384.65 ± 122.70
Incongruent long negative	371.41 ± 64.71	387.35 ± 108.81
Congruent short neutral	362.56 ± 52.04	389.85 ± 109.67
Incongruent short neutral	374.47 ± 64.99	396.40 ± 106.45
Congruent long neutral	376.66 ± 72.67	392.91 ± 135.40
Incongruent long neutral	360.44 ± 73.85	380.10 ± 90.39

anxiety, $F(1, 62) = 1.083$, $p = 0.302$, $\eta^2 = 0.017$, probe congruency $F(1, 62) = 1.882$, $p = 0.175$, $\eta^2 = 0.029$, presentation length $F(1, 62) = 3.833$, $p = 0.055$, $\eta^2 = 0.058$, or stimulus type $F(1, 62) = 0.532$, $p = 0.589$, $\eta^2 = 0.009$ were observed. Furthermore, no significant interactions were observed featuring math anxiety (all $p > 0.05$). For descriptive statistics please see Table 3 below.

Bivariate correlations revealed significant, weak positive correlations between math anxiety and RT in specific trial conditions: incongruent rumination-related short trials, $r(65) = 0.228$, $p = 0.04$, incongruent rumination-related long trials, $r(65) = 0.223$, $p = 0.04$, incongruent math-unrelated negative short trials, $r(65) = 0.256$, $p = 0.02$, and congruent neutral short trials, $r(65) = 0.246$, $p = .03$. All other correlations were non-significant ($p > 0.05$).

Finally, a bias index was calculated for both short and long trial types for rumination-related and math-unrelated negative stimuli. Specifically, we compared incongruent vs. baseline and congruent vs. baseline trials for both math-related and math-unrelated negative stimuli. To explore the role of math anxiety, independent sample t-tests were conducted comparing high and low math anxiety groups using quartiles. These analyses revealed no significant group differences in bias index scores for either short or long trial types ($p > 0.05$). Additionally, bivariate correlations between continuous math anxiety scores and bias index values revealed no significant relationships for either short or long trials in both rumination-related and math-unrelated negative stimuli categories (all $p > 0.05$).

General discussion

The present research aimed to investigate attentional biases (AB) in individuals with math anxiety (MA) using the dot-probe task, focusing on different content (rumination-related, negative, neutral) and presentation durations (500ms, 1250ms). Three studies were conducted to provide a nuanced understanding of AB in the context of MA. Contrary to our expectations, the results revealed a more complex picture of AB in individuals with MA.

Study 1 sought to examine whether individuals with high MA exhibit an attentional bias toward math-related stimuli, replicating previous research (Rubinsten et al., 2015; Suárez-Pellicioni et al., 2015). No significant differences in RT were observed between congruent and incongruent trials when comparing math anxiety groups, and no significant correlation was found between MA and the AB index. However, a positive correlation emerged between MA and RTs during congruent trials. This finding is somewhat counterintuitive, as we expected to see faster RTs, compatible with vigilance towards math content. This finding suggests attentional freezing, consistent with difficulty disengaging from math content, or inefficient processing, with math anxiety impairing participants' ability to respond efficiently when math-related stimuli are encountered (Suárez-Pellicioni et al., 2013). This contrasts with earlier reports of vigilance or avoidance effects in MA (Bar-Haim et al., 2007; Rubinsten et al., 2015), suggesting that methodological differences such a stimulus type (e.g. numerical equations or symbolic math problems versus word-based stimuli used here) or sample composition (e.g. more balanced or STEM-oriented cohorts in prior work versus predominantly female participants in the present study) may explain why our findings diverge.

Study 2 explored the temporal dynamics and content specificity of AB by including words associated with intrusive thoughts and ruminations that are common in math anxiety, measuring RTs across two exposure durations. Although no significant AB was found, we found a main effect of math anxiety with individuals with high MA exhibiting slower RTs across all trial types. Similarly, a positive correlation was found between math anxiety and all types of stimuli, regardless of content and presentation duration. These results are consistent with Study 1 showing slower RTs to be positively associated with math anxiety, although in this case it was only for congruent trials. These results indicate that MA is associated with a general processing delay or processing inefficiency rather than a specific attentional bias. This pattern is consistent with PET and ACT (Ashcraft & Kirk, 2001; González-Gómez et al., 2023; Shi & Liu, 2016), which propose that anxiety reduces the efficiency of cognitive performance by taxing attentional control and limited working memory resources.

Study 3 expanded the scope by including negative words unrelated to math to assess whether high MA individuals show a more generalised AB toward negative stimuli. Again, no significant effects emerged in RTs, probe congruency, or presentation length. However, weak positive correlations between MA and RTs for specific conditions such as incongruent math-related, math-unrelated negative and congruent neutral were observed. This suggests that individuals with higher math anxiety might show subtle attentional inefficiencies across different types of stimuli.

Taken together, these results suggest that MA may not manifest as a specific AB toward math-related or negative stimuli, but rather as a general slowing of cognitive processing. This slowing likely reflects lower processing efficiency (Suárez-Pellicioni et al., 2013), consistent with research indicating that math anxiety impairs task performance, particularly by consuming cognitive resources that are needed for task execution (Ashcraft & Kirk, 2001; González-Gómez et al., 2023; Shi & Liu, 2016). Our findings therefore add nuance to prevailing models such as the Vigilance-Avoidance Hypothesis (Derakshan et al., 2007), by suggesting that MA may impair general attentional control mechanisms rather than producing selective vigilance or avoidance of math stimuli.

The absence of a clear AB effect may also be due to methodological limitations. Previous studies have suggested that AB can be influenced by factors such as task-specific demands, stimulus complexity, or the emotional valence of stimuli (Bar-Haim et al., 2007; Schmukle, 2005; Xu et al., 2025). The lack of significant AB effects in this study might be due to the limited scope of the stimuli used. The word-based stimuli, while appropriate for assessing general attention, may not have sufficiently captured the cognitive and emotional responses associated with math anxiety. Studies incorporating more complex math-related tasks, such as numeric problems or word problems, have yielded different results (see Rubinsten et al., 2015). It is possible that symbolic or numeric stimuli evoke stronger affective responses than words alone, which may account for the absence of more robust AB effects in our study. Moreover, MA may have more subtle effects on attention that do not manifest through traditional RT measures but rather through other cognitive processes, such as working memory or task persistence. This issue may have been complicated by the online testing environment employed in studies 2 and 3, where RT measures can be noisier than laboratory-based environments, potentially obscuring more subtle attentional effects.

Another limitation of the dot probe task is that it offers a snapshot of attentional bias that is limited to the time during which the stimuli are presented. For example, if a study finds AB for a stimuli presented for 1000ms, nothing can be said about AB at shorter (500ms) or longer (2000ms) presentation durations, unless this specific time condition was tested (Mogg & Bradley, 1998). This is important because different results are likely to be obtained by testing different time frames (Bar-Haim et al., 2007). Therefore, the absence of attentional bias in the current study may, in part, be attributed to the specific stimulus presentation durations employed—500ms and 1250ms. By focusing on mid-to-late presentation windows, our study may have missed very early attentional effects, occurring sooner than 500ms. Studies in other anxiety disorders have reported evidence of vigilance for threat even for stimuli presented subliminally (Mogg & Bradley, 1999).

While existing research has generally emphasised a specific AB toward math-related stimuli in individuals with MA, our results suggest a broader cognitive effect characterised by slower processing speeds rather than a specific AB. Our results do not align neatly with prevailing theories like the Vigilance-Avoidance Hypothesis (Derakshan et al., 2007), which assumes that individuals with anxiety tend to show selective

avoidance or hypervigilance toward certain stimuli. Instead, our results suggest that the impact of MA may be more generalised, affecting processing efficiency rather than creating a specific bias toward or away from math-related stimuli. By proposing that MA leads to generalised cognitive slowing rather than specific ABs, this study adds an important dimension to the literature on MA. This interpretation extends current models by highlighting that MA may interfere with cognitive processing more broadly, aligning with theories that emphasise inefficiency of cognitive processing under anxiety, such as PET and ACT (Eysenck et al., 2007; González-Gómez et al., 2023).

The broader implications of this research are particularly relevant for developing interventions for individuals with MA. Whilst many existing interventions target emotion regulation and cognitive support (e.g. study skill and number sense training) in addressing MA (Sammallahti et al., 2023) the current study suggests that interventions targeting cognitive processing speed and overall attentional control could be worthwhile. For instance, interventions that improve cognitive efficiency - such as working memory training or mindfulness practices - could potentially help individuals with MA overcome generalised cognitive slowing or processing inefficiency by reducing intrusive thoughts and strengthening attentional regulation (Nazari et al., 2022; Shanmugarajah et al., 2025). Moreover, understanding that MA might not only cause a specific AB but rather a more generalised impact on cognitive processing could potentially inform educational interventions. For example, providing students with extra time, reducing time-pressure on math tasks, or incorporating structured breaks could potentially help mitigate the effects of processing inefficiency and enhance performance in students suffering from math anxiety.

Limitations and future research

Firstly, although the final sample size of 64 participants was sufficient for most planned analyses in Study 3, it was slightly below the recommended threshold for detecting medium-sized effects in a four-way mixed-design ANOVA. As a result, the study may have been underpowered to detect smaller or more complex interaction effects, particularly the higher-order interactions. Future studies should aim to recruit larger samples to ensure adequate sensitivity for such designs.

Secondly, whilst widely used in anxiety research and considered more ecologically valid than the Emotional Stroop task, the dot-probe task may not be the optimal method for capturing AB in MA. For example, Chapman et al. (2019) found that while the dot-probe task can detect early attentional engagement (around 100ms), its ability to measure sustained AB is limited. This limitation is reflected in its poor internal reliability, particularly at longer stimulus onset asynchronies, which may have contributed to the failure to capture the complexity of AB in our studies. Given concerns about the internal reliability of the dot-probe task (Waechter et al., 2014; Xu et al., 2025) it is possible that standard subtraction-based RT or accuracy index scores are unreliable for assessing AB to threat, regardless of anxiety status.

To address the limitations of the dot-probe task, future research should consider alternative methods for assessing AB in MA. Eye-tracking presents a promising tool for capturing more continuous and reliable measures of attentional deployment, enabling a more detailed analysis of how attention shifts over time (Gaffiero et al., 2019). Eye-tracking would allow researchers to examine both early engagement and sustained attention, offering a more comprehensive understanding of how AB unfolds in individuals with MA. Indeed, existing eye-tracking studies have provided useful insight into general eye-movement behaviour (Hunt et al., 2014) and pupil dilation (Galeano & Gredebäck, 2024) as a function of MA. Additionally, incorporating more complex math-related stimuli, such as math problems or word problems, could provide better insight into how individuals with high MA respond to math-specific tasks. Previous research suggests that more complex stimuli elicit clearer ABs in math anxiety (Rubinsten et al., 2015), indicating that the current word-based approach may not fully capture the attentional responses associated with math anxiety. After all, participants in our studies viewed words, not numbers or formulas that are more strongly associated with this subject. In addition to eye-tracking, EEG-based measures could provide valuable insight into the temporal dynamics of attentional allocation in MA (Batashvili et al., 2019). For example, Event Related Potentials (ERPs) allow for more precise measurement of early vs. late attentional processes (Suárez-Pellicioni et al., 2015) and could complement data obtained via eye-tracking technology.

Thirdly, the three studies employed different participant sources. This recruitment strategy reflected practical constraints, such as availability of undergraduate participants for Study 1, and broader online access to participants for Studies 2 and 3, while also allowing us to examine attentional processes across more diverse samples, thereby enhancing the generalisability of our findings. Nevertheless, variability in age, educational background, academic subject and math familiarity across samples may have influenced the results, and future research should aim to standardise recruitment or compare these groups directly. In addition, the gender distribution was skewed towards female participants, which is important to consider given evidence that women typically report higher levels of math anxiety (Devine et al., 2012). Results should therefore be interpreted with this limitation in mind.

Fourthly, although the use of median and quartile splits to classify participants into high and low MA groups is consistent with previous research (Bar-Haim et al., 2007), it is inherently a data-driven approach that may be less robust in smaller samples. Such procedures can lead to reduced statistical power, artificial dichotomisation of continuous variation, and potential misclassification of participants (McClelland et al., 2015). While this approach was adopted here to facilitate interpretability, future research with larger samples should prioritise using mathematics anxiety as a continuous variable.

Finally, future studies should aim to diversify participant samples to enhance the generalisability of findings. Including individuals from various demographic backgrounds and educational levels would provide a more comprehensive understanding of how MA manifests across different populations. Since MA is prevalent across a range of educational contexts, examining how it influences attentional processes in more diverse groups would help refine theoretical models and inform the development of interventions tailored to meet the needs of varied populations. Moreover, applying EEG or eye-tracking approaches in more diverse participant groups would provide both greater ecological validity and a richer understanding of how MA influences attentional processes across different contexts.

Conclusion

Our study found no evidence of an AB to math-related, rumination-related or math-unrelated negative words in math anxiety. Instead, we found that MA is associated with a general slowing of cognitive processing, suggesting lack of processing efficiency. This may indicate that other mechanisms, such as working memory interference, avoidance, or cognitive control deficits may play a more critical role in math anxiety. To better understand the mechanisms underlying MA, future research should incorporate more nuanced measures, such as eye-tracking and explore more complex stimuli more strongly associated with math, such as numeric expressions, formulas, equations, etc. While the dot-probe task remains a widely used tool in attentional bias research, its limitations in capturing the full complexity of attentional processes suggest that alternative methodologies are needed. By addressing the methodological challenges identified in these studies, future research can more effectively investigate how MA affects cognitive processes and develop targeted interventions to mitigate its impact.

Author contributions

CRedit: Conceptualization: DG, TEH; Data curation: DG; Formal Analysis: DG, TEH; Investigation: DG; Methodology: DG, TEH; Project administration: TEH; Resources: DG, TEH; Supervision: MSP, TEH; Validation: DG, MSP, TEH; Visualization: DG, MSP, TEH; Writing – original draft: DG, MSP, TEH; Writing – review & editing: DG, MSP, TEH.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethical approval

Study 1 was conducted as part of an undergraduate dissertation project at the University of Derby. In line with institutional policy, undergraduate projects were reviewed through the School of Psychology ethics procedures but were not assigned individual reference numbers. Study 2 was reviewed and approved by the University of Derby

College of Health, Psychology and Social Care Research Ethics Committee (reference: ETH2021-2024). Study 3 was reviewed and approved by the University of Derby College of Health, Psychology and Social Care Research Ethics Committee (reference: ETH2122-3192). All studies were carried out in accordance with the ethical standards of the British Psychological Society and the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to participation.

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Data availability statement

Data is available upon request.

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